Norms of Error Terms in Transformer Language Models Predict Human Reading Time Beyond Surprisal

Ziyuan Cao[1], William Schuler[1]
[1]Ohio State University

Surprisal theory [3, 5] proposes that the processing difficulty of a word in incremental sentence processing is proportional to the negative log-probability of that word given the context. Recent work proposed variants of surprisal that better model human sentence processing [1, 11, 7]. [7] investigates similarity-adjusted surprisal where it aggregates the probability mass of all the words in the vocabulary weighted by their similarity to the target word. Similarly inspired, the processing difficulties incurred by seeing a word may be correlated to the similarity between the representation of the context and an alternative representation that would have made the observed word more predictable. We measure this using the norm of the error term back-propagated from the surprisal signal to different layers inside a neural language model. We observe that the L2 norm of the error terms at the input of the final output embedding of GPT-2 [10] is closely related to the similarity-adjusted surprisal [7] when using GPT-2 embedding-based cosine similarity as the similarity function. This allows us to generalize similarity-adjusted surprisal by looking at the norm of error terms propagated more deeply into the network, i.e., $\nabla_{h_l} \| -\log p_t \|_2$, for some intermediate layer h_l. This also has connection to error-based psycholinguistic predictors derived from predictive coding network [9] as it was shown that predictive coding algorithm converges to the error back-propagation algorithm of artificial neural network under certain conditions [13, 8].

We evaluate the psycholinguistic predictive power of the norm of the error terms of GPT-2 with regression experiments on multiple reading time datasets. We conduct regression experiments on both self-paced reading (SPR) and eye-tracking (ET) datasets, including Natural Stories [2], Brown [12], Dundee [4], and Provo [6]. We first fit a linear mixed effect regression model with baseline predictors. Then we fit another model with the norm of the error term as an additional predictor. For SPR datasets, we include word position, word length, and GPT-2 small surprisal as the baseline predictors. For ET datasets, we include word position, word length, saccade distance in word, whether the previous word is fixated, and GPT-2 small surprisal as the baseline predictors. We evaluate whether the norm of the error terms provides predictive power beyond the baseline predictors by examining the difference in log-likelihood between the baseline model and the full model using 10-fold cross validation. As shown in Table 1, the norm of error terms at layer 5 contributes to reading time prediction in addition to baseline predictors. Permutation tests across the 10 folds show that the improvement is statistically significant.

We additionally evaluate the predictive power of the norms of the error terms at all the layers for the four datasets. The experiment setup is the same except that we did not include spillover predictors. The results are shown in Figure 1. Layers 0-11 represent the error term at the output of each transformer module. Layer 12 means the error term at the input of the final output embedding. Predictive power across layers on the four datasets shows a similar trend. It shows the strongest predictive power at layer 0, i.e., the output of the first transformer module, and reaches a lower peak around layer 10. It suggests that human processing difficulties may be particularly sensitive to the error terms propagated to the representation of the local preceding context and to some high-level representation of the context. These results are consistent with the finding that dynamics-based predictors derived from predictive coding network reliably predict human RTs[9].

	Δ Log-likelihood
Natural Stories	637.4^{***}
Brown	158.8^{***}
Dundee	162.9^{***}
Provo	37.5^{***}

Table 1: Δ Log-likelihood of adding the norm of the error terms at layer 5 of GPT-2 small as predictors of human RTs over surprisal baseline. All predictors at the previous position except word position are included to account for spillover.

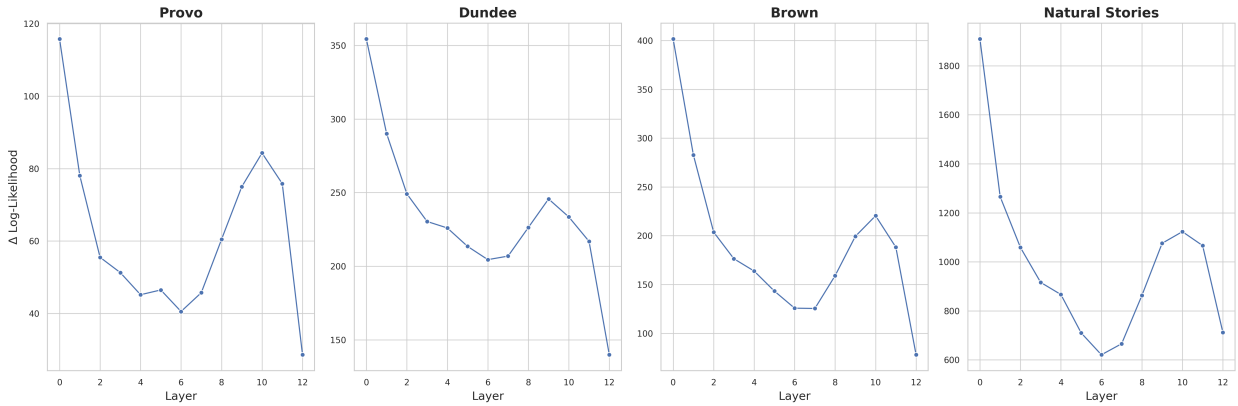

Figure 1: Δ Log-likelihood of adding the norm of the error terms at different layers of GPT-2 small as predictors of human RTs over surprisal baseline (no spillover). Layer 0 means the output of the first transformer module. Layer 11 means the output of the last transformer module. Layer 12 means the output of the final layer normalization, i.e., the input of the output embedding.

References

[1] Richard Futrell and Roger Levy. In: *EACL 2017*. Valencia, Spain, Apr. 2017.

[2] Richard Futrell et al. In: *LREC 2018*. Miyazaki, Japan, May 2018.

[3] John Hale. In: *NAACL*. 2001.

[4] Alan Kennedy, Robin Hill, and Joël Pynte. In: *Proceedings of the 12th European conference on eye movement*. 2003.

[5] Roger Levy. In: *Cognition* (2008).

[6] Steven G Luke and Kiel Christianson. In: *Behavior research methods* (2018).

[7] Clara Meister, Mario Giulianelli, and Tiago Pimentel. In: *EMNLP 2024*. Miami, Florida, USA, Nov. 2024.

[8] Beren Millidge, Alexander Tschantz, and Christopher L Buckley. In: *Neural Computation* (2022).

[9] Chiebuka Ohams et al. In: *Journal of Memory and Language* (2026).

[10] Alec Radford et al. In: *OpenAI blog* (2019).

[11] Marten van Schijndel and Tal Linzen. In: *EMNLP 2018*. Brussels, Belgium, Oct. 2018.

[12] Nathaniel J Smith and Roger Levy. In: *Cognition* (2013).

[13] James CR Whittington and Rafal Bogacz. In: *Neural computation* (2017).